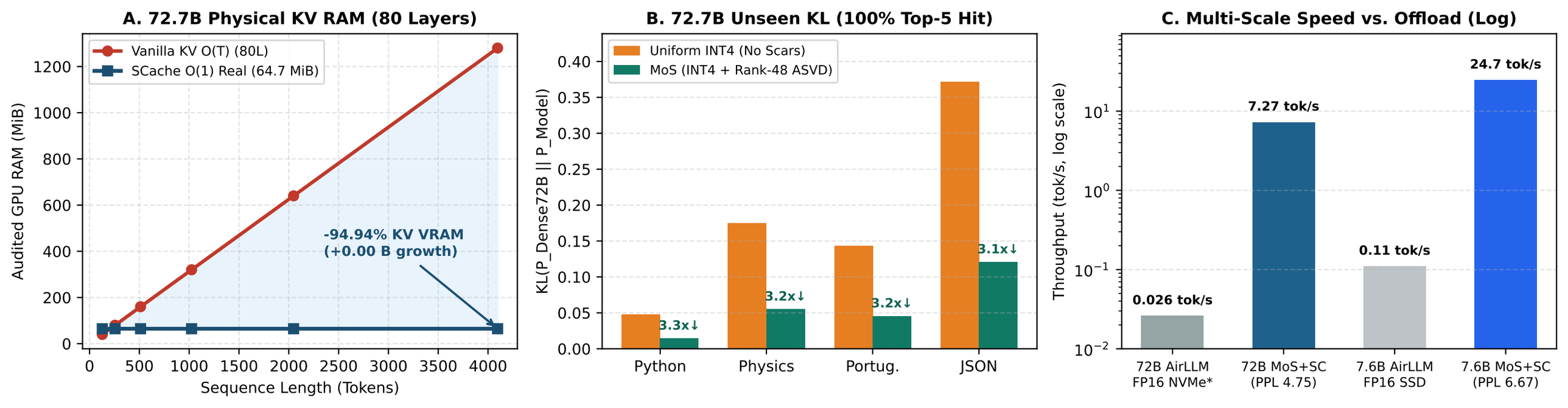
A. 72.7B Physical KV RAM (80 Layers)
Vanilla KV O(T) (80L)
SCache O(1) Real (64.7 MiB)
-94.94% KV VRAM
(+0.00 B growth)
Audited GPU RAM (MiB)
Sequence Length (Tokens)
0
200
400
600
800
1000
1200
0
500
1000
1500
2000
2500
3000
3500
4000
B. 72.7B Unseen KL (100% Top-5 Hit)
Uniform INT4 (No Scars)
MoS (INT4 + Rank-48 ASVD)
KL(P_Dense72B || P_Model)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
3.3x↓
3.2x↓
3.2x↓
3.1x↓
Python
Physics
Portug.
JSON
C. Multi-Scale Speed vs. Offload (Log)
Throughput (tok/s, log scale)
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
0.026 tok/s
7.27 tok/s
0.11 tok/s
24.7 tok/s
72B AirLLM
FP16 NVMe*
72B MoS+SC
(PPL 4.75)
7.6B AirLLM
FP16 SSD
7.6B MoS+SC
(PPL 6.67)